MIL-H-27269A(USAF)
20 December 1977
SUPERSEDING
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31 December 1959

MILITARY SPECIFICATION

HORIZONTAL SITUATION INDICATOR GROUP AF/A24J-1

This specification is approved for use by Code 99, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for a servo driven course-bearing-heading-situation indicator, designated AF/A24J-1.

*2. APPLICABLE DOCUMENTS

2.1 Issues of Documents. The following documents of the issue in effect on date of invitation for bids of request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-636	Box, Fiberboard
QQ-P-416	Plating, Cadmium (Electrodeposited)

MILITARY

MIL-P-116	Preservation, Method of
MIL-C-675	Coating of Glass Optical Elements (Anti-reflection)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: (Hq AFLC CASO/LODS, Federal Center, Battle Creek MI 49016) by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6610

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MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Aeronautical
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-L-25467	Lighting, Integral, Instrument, General Specification For

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-838	Lubrication of Military Equipment
MIL-STD-889	Dissimilar Metals
MS33545	Case - Instrument 5 x 5-1/4 Standard Dimensions For
MS33558	Numerials and Letters, Aircraft Instrument Dial, Standard Form of

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 QUALIFICATION. The horizontal situation indicator group furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein.

3.2 COMPONENTS. The horizontal situation indicator group shall consist of an indicator and remote unit consisting of four servo amplifiers and associated power supplies.

*3.3 SELECTION OF SPECIFICATIONS AND STANDARDS. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 MATERIALS

3.4.1 FUNGUS-PROOF MATERIALS. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.4.2 NONMAGNETIC MATERIALS. Nonmagnetic materials shall be used for all parts except where magnetic materials are essential.

3.4.3 NONFERROUS MATERIALS. Nonferrous materials shall be used for all parts of the horizontal situation indicator group except where ferrous materials are essential.

3.4.4 METALS. Metals shall be of the corrosion-resistant type or shall be suitably protected to resist corrosion during normal service life. Materials contained within hermetically sealed inclosures are considered to be suitably protected from corrosion. Requirements specified for fungicidal and corrosion protective treatment, anodizing of the aluminum alloy parts and cadmium plating of steel parts are not applicable for parts within hermetically sealed instruments.

*3.4.5 DISSIMILAR METALS. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

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3.4.6 TOXIC AND CORROSIVE FUMES. The materials, as installed in the horizontal situation indicator group shall not liberate deleterious or corrosive fumes. This shall include any fungicidal agents that are used.

3.4.7 PROTECTIVE TREATMENT. When materials are used in the construction of the horizontal situation indicator group that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5 DESIGN AND CONSTRUCTION

*3.5.1 The horizontal situation indicator group shall be designed to present the following information:

a. Stabilized magnetic compass heading provided from the master directional reference.

b. Command heading information provided from a remote source or desired heading set in manually.

c. Command course provided from a remote source or desired course set in manually displayed by a pointer and a numerical indication.

d. Course bar which indicates displacement of the aircraft from a manually or automatically selected track.

e. To-from indication from a selected radio facility.

f. Bearing to station or target.

g. Numerical indication of distance to a transmitting radio source or target; indication masked when not in use.

h. Mode words internally lighted on instrument display.

i. Bearing from a station or target (reciprocal bearing).

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5.2 The horizontal situation indicator group shall be so con-:ructed that no parts will work loose in service. It shall be ıilt to withstand the strains, jars, vibrations, and other con-.tions incident to shipping, storage, installation, and service.

5.3 The horizontal situation indicator group shall be so con-:ructed that overhaul, including replacement of worn parts, ljustment balance, and calibration may be accomplished by ıstrument and electronic repairman provided with simplified ›ols and equipment, and trained in the repair of electronic ¡uipment.

5.4 MINIATURIZATION. Modern techniques of miniaturization, ›tting, and hermetic sealing shall be exploited to the greatest ttent practicable without sacrificing ruggedness, reliability, ıd service life.

5.5 TRANSISTORIZED OR MAGNETIC AMPLIFIER CIRCUITS. Four :ansistorized or magnetic servo amplifiers shall be housed in a :parate package from the indicator proper. The amplifiers ıall feature module plug-in capability. Each amplifier module ıall be designed with partial redundant circuits so that the ;rvos will continue to operate though with reduced performance, ıould partial failure of the amplifiers occur.

.6 PERFORMANCE. The horizontal situation indicator group ıall be capable of satisfactory operation when subjected to the ›llowing conditions:

a. Temperature. Temperatures over the range of -54° to 71°C. The remote amplifier shall be operable for 10 minutes at 98°C. Storage at temperatures between -65° and +85°C shall not ause any detrimental effects.

b. Altitude. All altitudes from sea level to 100,000 feet.

c. Humidity. Relative humidity up to 100 percent.

d. Fungus. Fungus growth as encountered in tropical cli-ıtes.

e. Salt Spray. Exposure to salt laden atmosphere.

f. Sand and Dust. Sand and dust particles as encountered ı desert areas.

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g. Thermal Shock. Thermal shock forces as encountered during rapid changes of extreme temperature.

h. Vibration. Vibration forces incident to service use.

i. Sealing. Immersion in water with pressure applied, or leak test conducted with an approved helium leak detector, without evidence of leakage in excess of that specified herein.

j. Acceleration. Acceleration forces as encountered in high performance aircraft.

k. Attitude. The horizontal situation indicator group shall be capable of operating satisfactorily while in any attitude.

l. Friction Error. A minimum of error shall be introduced in the readings due to the friction between moving parts.

m. Magnetic Effect. The horizontal situation indicator group shall not cause excessive error in the indication of panel-mounted directional magnetic compasses.

n. Life. The indicator shall be capable of meeting the performance requirements for 1000-hour operation during actual service usage. This is to be demonstrated in the 1000-hour life test in 4.4.24.

*3.6.1 ELECTROMAGNETIC INTERFERENCE. The equipment shall meet the electromagnetic interference requirements of MIL-STD-461 for class ID.

*3.7 INTERCHANGEABILITY. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-STD-100 shall govern changes in the manufacturer's part numbers.

*3.8 INDICATOR CASE. The indicator case shall be in accordance with MS33545 with the following exceptions:

a. Dimensions shall be as shown on figure 1.

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b. Electrical receptacles specified herein shall be attached o the indicator case. Forces of the magnitude shown in figure 1 pplied normally to the elctrical connectors or to the bezel in jecting the indicator shall not damage the case.

c. The dimensions of the cover glass aperture shall be as hown on figure 2.

d. Removable guidepins as shown on figure 1 shall be pro- ·ided at the rear of the case.

e. Control knobs as shown on figure 2, shall be provided. he inside of the case around the connectors shall be of suffi- :ient strength to prevent damage to the case when the connectors ıre tightened during installation. The case shal' be made of ıetal, uniform in texture with a smooth surface and entirely :overed with a durable dull black finish. The control knobs ıhall be color No. 32631 of FED-STD-595, the end of the knobs ıhall be black with white lettering. A minimum of 2 ounce- .nches and a maximum of 10 ounce-inches shall be required to turn :he knobs. Sealed fluid couplings may be utilized for the con- :rol knobs.

3.8.1 COVER GLASS. The cover glass shall be hermetically ıealed with the indicator case. It shall be mounted in the flange ›y inserting it from within the case. The cover glass shall be ›etween 3/16 inch to 1/4 inch in thickness. It shall be flat and :ree from discoloration (except as induced by glare reducing :oatings) scratches and striae which interfere with normal reading ›f the indicator. The cover glass aperture shall be as specified ›n figure 2.

*3.8.1.1 ANTIREFLECTION COATING. A magnesium fluoride coating in accordance with MIL-C-675 shall be applied to each surface of the cover glass and the lighting wedge. Coated surfaces shall ıot chip, scale, peel, flake, dissolve, discolor or otherwise be adversely affected in normal service use. The coating on the ›uter surface of the cover glass shall be capable of withstanding ıbrasion incident to normal service usage.

*3.8.2 INDICATOR DISPLAY. The indicator display shall conform to figure 2. All sizes of the pointers, azimuth card, deviation bar, numerals, and letters shall conform to the size and width indicated on figure 2. The azimuth card, main mask of the dis- play, course arrow, and deviation bar shall all lie on the same

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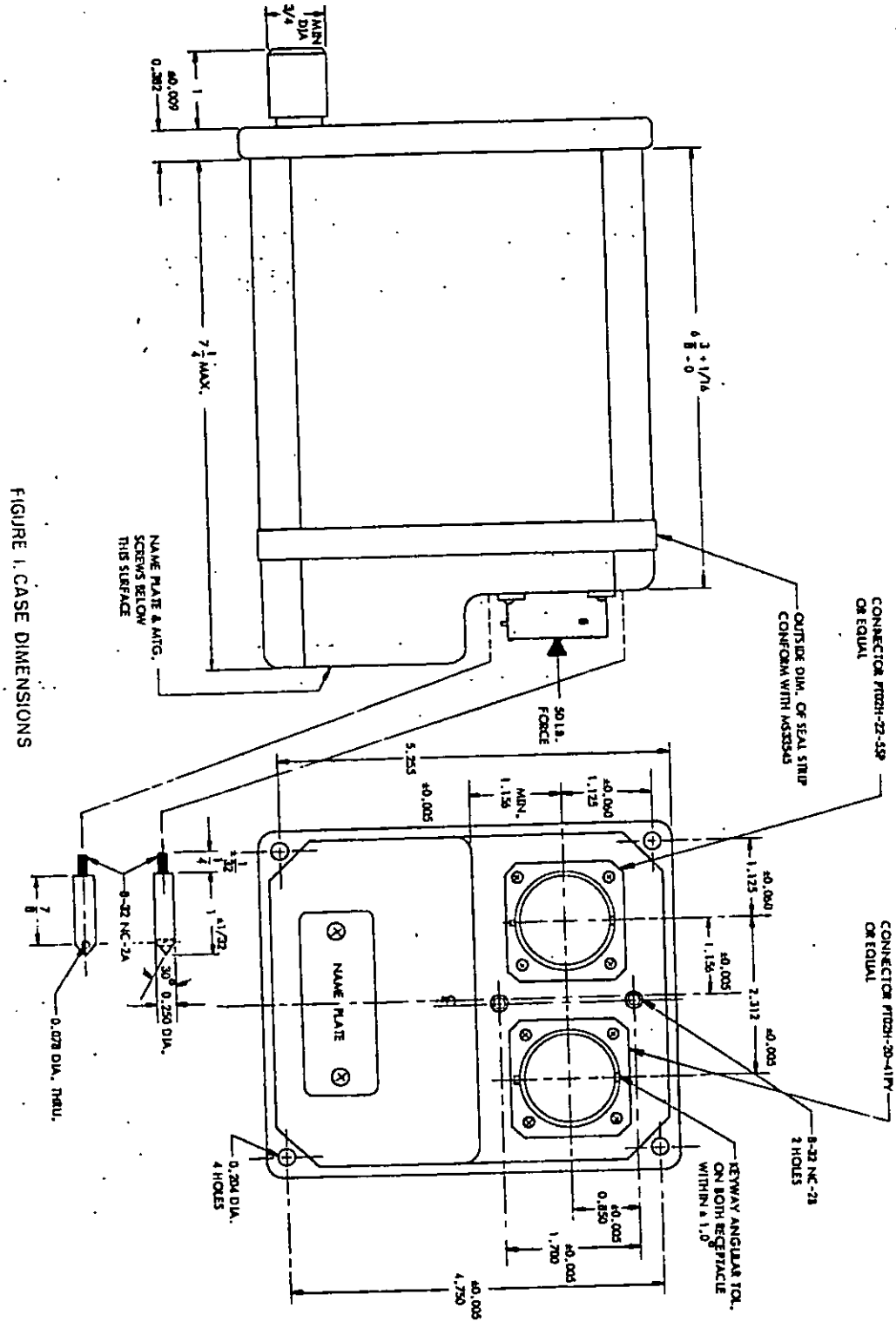
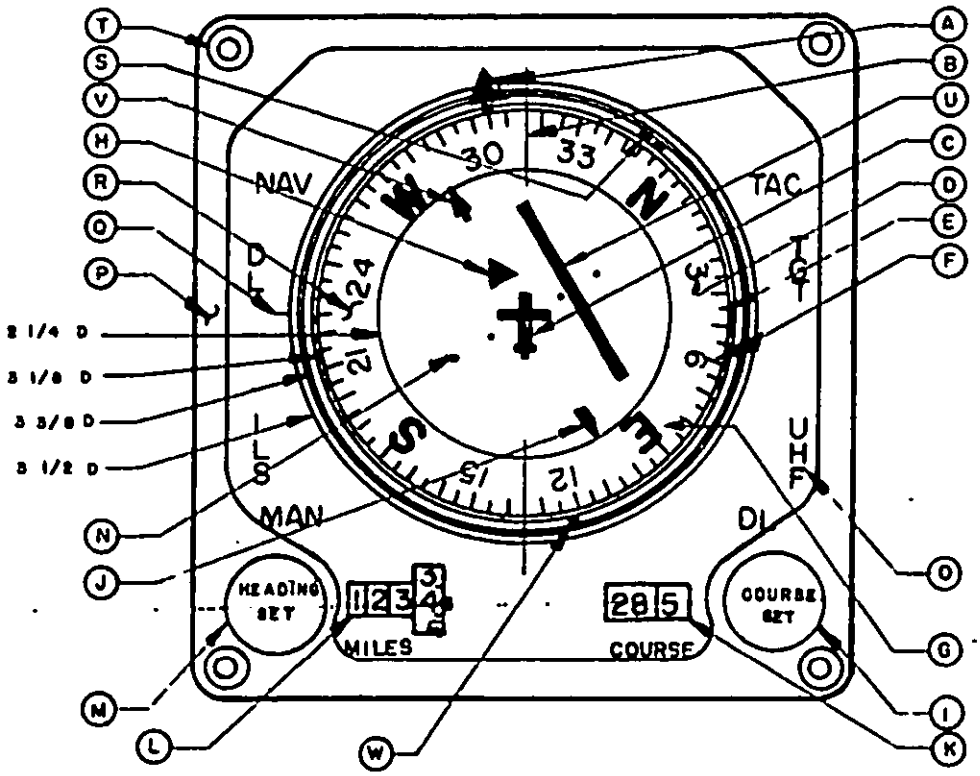


FIGURE 1. CASE DIMENSIONS

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A	BEARING POINTER	L	DISTANCE DISPLAY
B	LUBBER LINE	M	HEADING SELECTION KNOB
C	AIRPLANE SYMBOL	N	COURSE DEVIATION DOT
D	NUMERALS	O	MODE OF OPERATION WINDOW
E	5° INCREMENTS	P	CASE FLANGE
F	10° INCREMENTS	Q	FIXED MARKER
G	CARDINAL LETTERS	R	INDICATOR DIAL
H	TO-FROM ARROW	S	HEADING MARKER
I	COURSE SELECTION KNOB	T	MOUNTING HOLE
J	RECIPROCAL COURSE INDEX	U	COURSE BAR
K	SELECTED COURSE COUNTER	V	COURSE ARROW
		W	RECIPROCAL BEARING POINTER

DIMENSIONS IN INCHES

FIGURE 2. INDICATOR DISPLAY

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plane designated plane A and shall be 0.5 ±0.05 inch from the forward edge of the cover glass. The airplane and lubber line shall be 0.05 inch forward of plane A. The forward face of the bearing and reciprocal bearing pointer shall be 0.03 inch forward of plane A. The forward face of the heading marker shall be 0.01 inch forward of plane A. The first set of deviation dots shall be 0.04 inch back of plane A. The forward face of the to-from arrow shall be 0.12 inch back of plane A. The second set of deviation dots shall be 0.13 inch back of plane A. The deepest point of any visible portion of the display shall be 0.17 inch from plane A. These dimensions as given are design center values.

3.8.3 FUNCTION WORDS. Internally lighted function words shall be incorporated in the display. The size and position of the words shall be as specified on figure 2. Each word shall be illuminated with 2 lamps. The function words shall be as those shown on figure 2.

3.8.3.1 The function words shall be illuminated by 6V lamps. Each function word will have one common electrical connection and one specific pin connection to connector PT02H-22-55P, or equal. With 5V applied to the lamps the average intensity of the function words shall be 100-400 foot-lamberts. With no voltage applied and viewed in sunlight the function words shall not appear lighted.

*3.8.4 HERMETIC SEALING. The indicator case shall be hermetically sealed. The design of the case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed. This shall be accomplished without the use of special tools and fixtures unless they are approved by the procuring activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials that will be affected by the action of any atmosphere to which the component may be subjected. The case shall be so designed that it will meet the pressure requirements necessary when tested for leaks by means of a mass spectrometer type of helium leak detector.

3.8.4.1 FILLING MEDIUM. The filling medium shall be of at least 98 percent purity, free of dust particles and containing not more than 0.006 milligrams of water vapor per liter (dew point -65°C) at the filling pressure. The filling medium shall be either 100

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ırcent helium or a mixutre of 88 to 92 percent nitrogen and the ımainder helium. The absolute pressure of the filling medium ı the case shall be approximately 1 atmosphere. Where practiıble the 100 percent helium filling medium shall be utilized.

8.4.2 LEAK RATE. The indicator shall be sealed so that the ıitial maximum leak rate at a pressure differential of approxiıtley 1 atmosphere will not permit more than 10 percent loss of ıe total filling medium after 1000 hours.

8.5 LIGHTING. The indicator shall be integrally lighted in :cordance with MIL-L-25467. The lighting design shall employ a ıdge for transmission. The lighting shall be subject to approval ' the procuring activity. The integral lighting shall be comırable to and in brightness balance with a prime standard indiıtor. The prime standard shall be a horizontal situation indiıtor group indicator conforming to the requirements of the speci.cation, or lighted model of such an indicator. The prime :andard shall have applied to its lighting circuits the speci.c voltage that the procuring activity has established as being :cessary to assure equality of presentation (from a lighting :andpoint) with adjacent but dissimilar instruments. A resistor ıall be incorporated in series with the integral lighting cirıits within the case. The resistor shall be utilized to provide :act lighting balance with the prime standard.

9 ELECTRONIC COMPONENTS. Electronic components of the ›rizontal situation indicator group shall meet the applicable :quirements of MIL-E-5400 and as specified herein. In the event ıe requirements of MIL-E-5400 and this specification conflict, ıe requirements of this specification shall govern.

10 SERVO AMPLIFIER. The servo amplifier shall be housed in a :parate package and mounted remotely. The amplifier case :mensions shall be in accordance with figure 3. The design of ıe amplifier shall be such that it will meet the requirements : vibration test procedure I of MIL-E-5272 when mounted directly › the airframe without inclusion of a shockmount. The amplifer modules need not be hermetically sealed but shall be sealed ı such a manner that they shall withstand all of the environıntal conditions and environmental tests of this specification. ıe functional operation and power inputs to the amplifier unit ıall be in accordance with figure 4.

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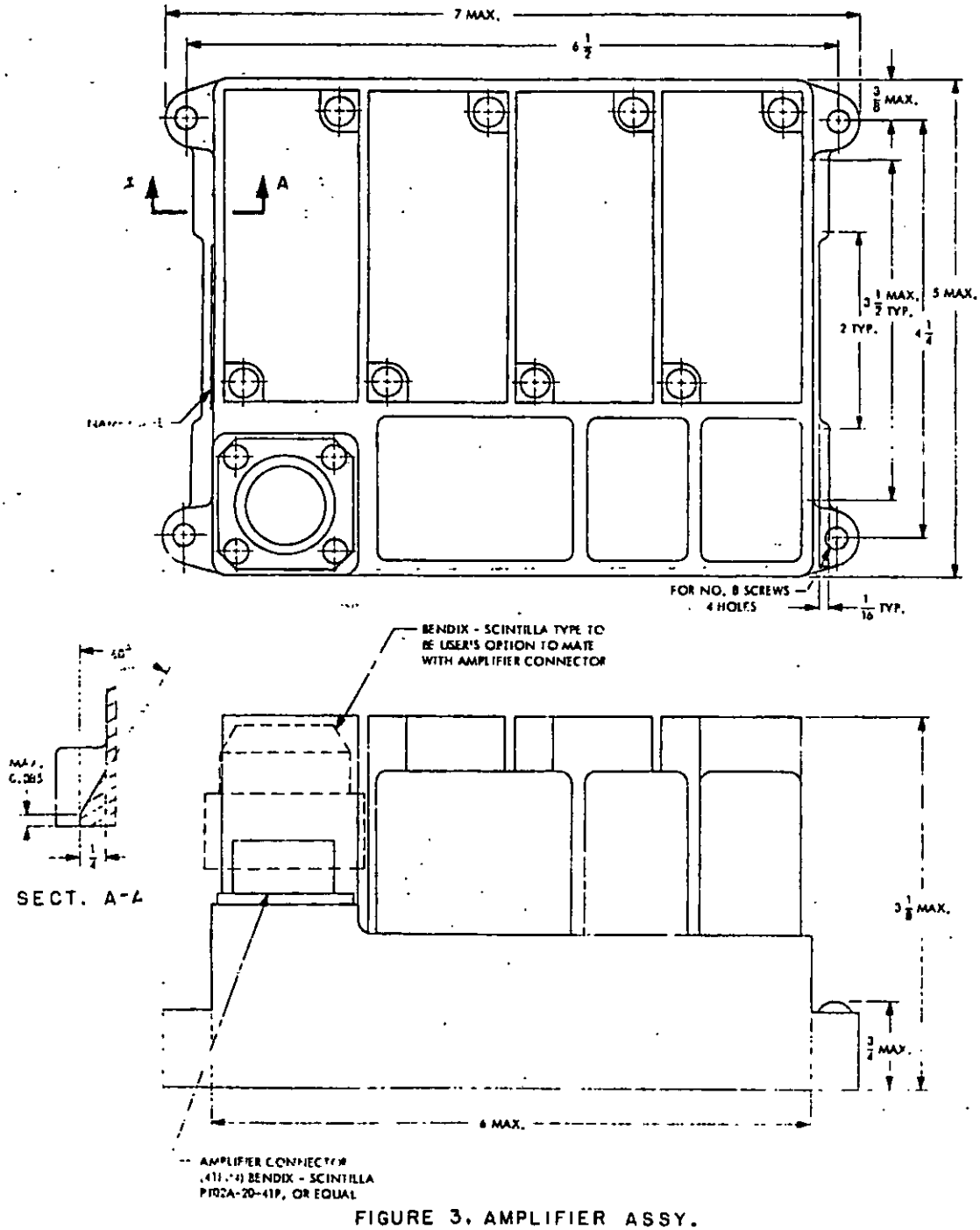


FIGURE 3. AMPLIFIER ASSY.

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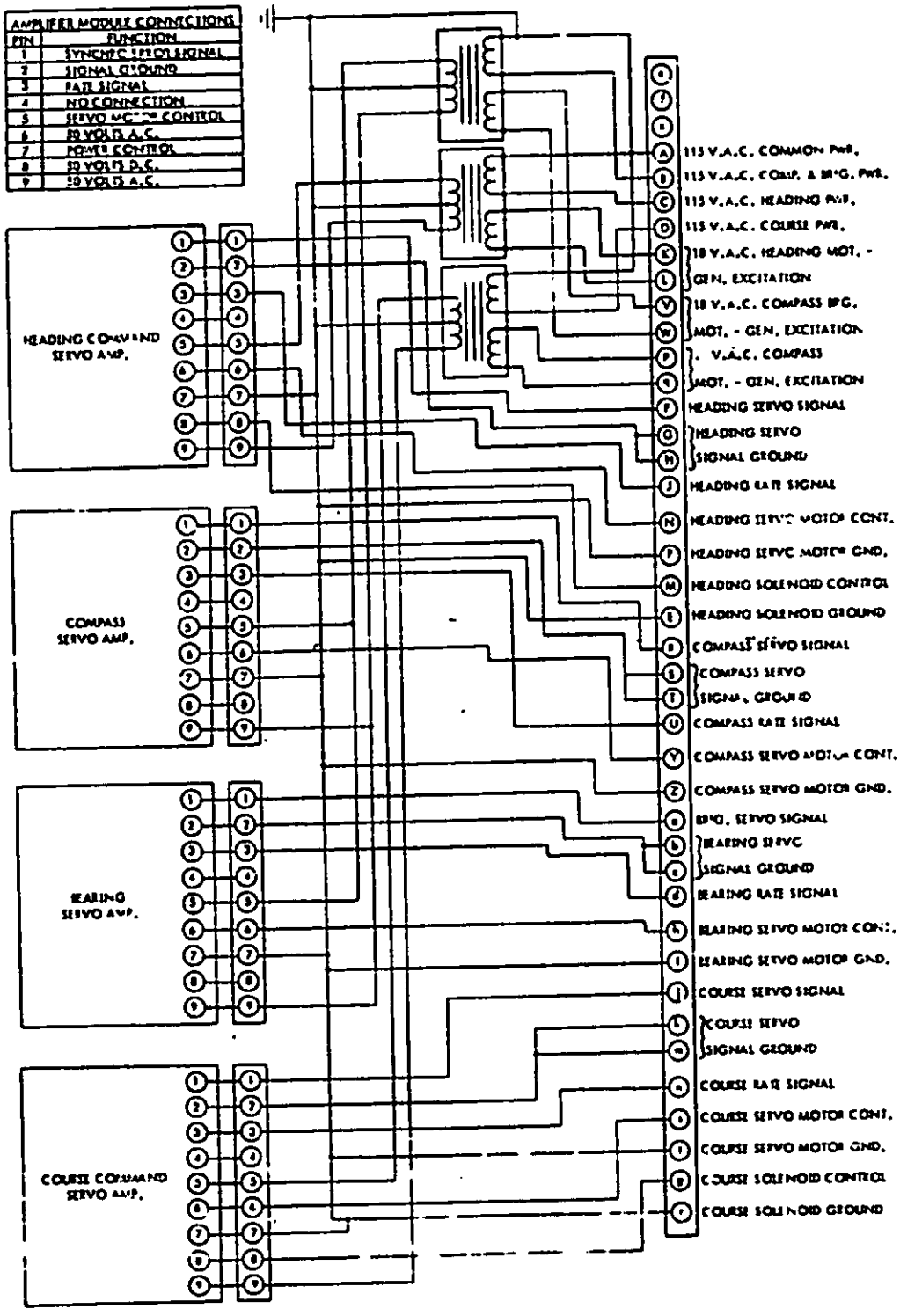


FIGURE 4. FUNCTIONAL SCHEMATIC, HSI REMOTE AMPLIFIER

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3.11 STANDARD SYNCHRO TEST TRANSMITTER. A standard calibrated synchro with minimum and known errors shall be an Eclipse-Pioneer Type AY201-1, or equal, high precision transmitting synchro. A 0° to 360° dial clamped to the rotor of the synchro shall be settable, rotatable, and readable through 360° of rotation to 6 minutes of arc. The following procedure shall be used to calibrate the test synchro.

3.11.1 With the rotor leads designated as H and C and with one stator lead designated as Z and connected to C. When 26V is applied to H and C (C being ground) the dial shall be positioned on 0° and clamped to the rotor when

a. The voltage across H and Y is maximum

b. The voltage across the remaining two stator leads (X and Y) is minimum.

c. The voltage across X and Z increases before it decreases for increasing heading indication of the dial. The voltage X to Z and Y to Z shall be in phase with the excitation voltage C to H when the dial reading is zero. For increasing heading indication of the dial the voltage X to Y shall increase and be in phase with the excitation voltage C to H. The test transmitter will be set at an index reference of zero and a positive rotation reference XYZ.

3.12 PRESENTATION. The presentation shall be in accordance with figure 2 and shall consist of a rotating-card type azimuth ring, lubber line, a heading marker, a course arrow, a to-from arrow, a course bar, a bearing pointer, a selected course window, a distance window, function words, and a miniature airplane symbol behind the cover glass of the instrument to present a pictorial display of the navigation situation.

3.13 LUBBER LINE AND AIRPLANE. The lubber line and airplane shall be in accordance with figure 2.

3.14 AIRCRAFT HEADING. The aircraft heading shall be displayed on the azimuth ring and shall be read under the fixed lubber line located at the top of the indicator face as shown on figure 2. The azimuth ring shall be in accordance with figure 2.

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3.14.1 AZIMUTH RING SIGNAL INPUT. The input synchro control transformer shall be a high impedance type EP AY500-5, or equal, which shall receive an input from a synchro device similar to the test transmitter described in 3.11. A size 8 synchro control transformer with the equal impedance of the stator windings as a Type EP AY500-5, or equal, can be utilized.

3.14.2 AZIMUTH RING SERVO SYSTEM. A servo follow-up system consisting of a control transformer, motor, rate generator feedback, and amplifier shall drive the azimuth card and rotor of the input control transformer to a null. It shall be capable of receiving a change of heading at any rate between 0° and 60° per second without hunting. There shall be no appreciable hunt or oscillation in the servo system while the input compass signal is changing or stationary. The servo system shall position the heading indication, under the lubber line, to within $\pm 1/2^{\circ}$ of the position of a calibrated synchro. The design of the azimuth card gearing shall result in operation as would be achieved in the system shown on figure 5 and the excitation voltage of the test transmitter C to H in phase with the power input pin A to pin B on the amplifier connector:

a. The azimuth ring shall position at 0° (north)when the rotor of the test transmitter dial is at 0°.

b. If the dial on the rotor of the test transmitter is continuously rotated so that its heading indication increases, the azimuth ring heading shall increase ccw (counterclockwise) rotation accordingly and the voltage C to H of the control transformer shall be in phase with the excitation voltage C to H.

3.15 COMMAND HEADING. A means of selecting a desired heading both automatically and manually shall be provided. A heading marker shown on figure 2, which can rotate around the azimuth ring, shall be driven by a servo system when excited by a remote device similar to the standard test transmitter specified in 3.11 and manually by a knob at the lower left corner of the instrument. There shall be no visible interaction between the knob and the servo motor so that when the remote synchro device is driving the servo system, turning of the knob should have no lasting effect on the heading marker position. The heading marker shall rotate in synchronism with the azimuth ring after it is set. The heading selection knob shall be connected with the heading marker in such a manner that cw (clockwise) rotation of the knob produces cw rotation of the heading marker, and vice versa. The turning ratio between the knob and the heading

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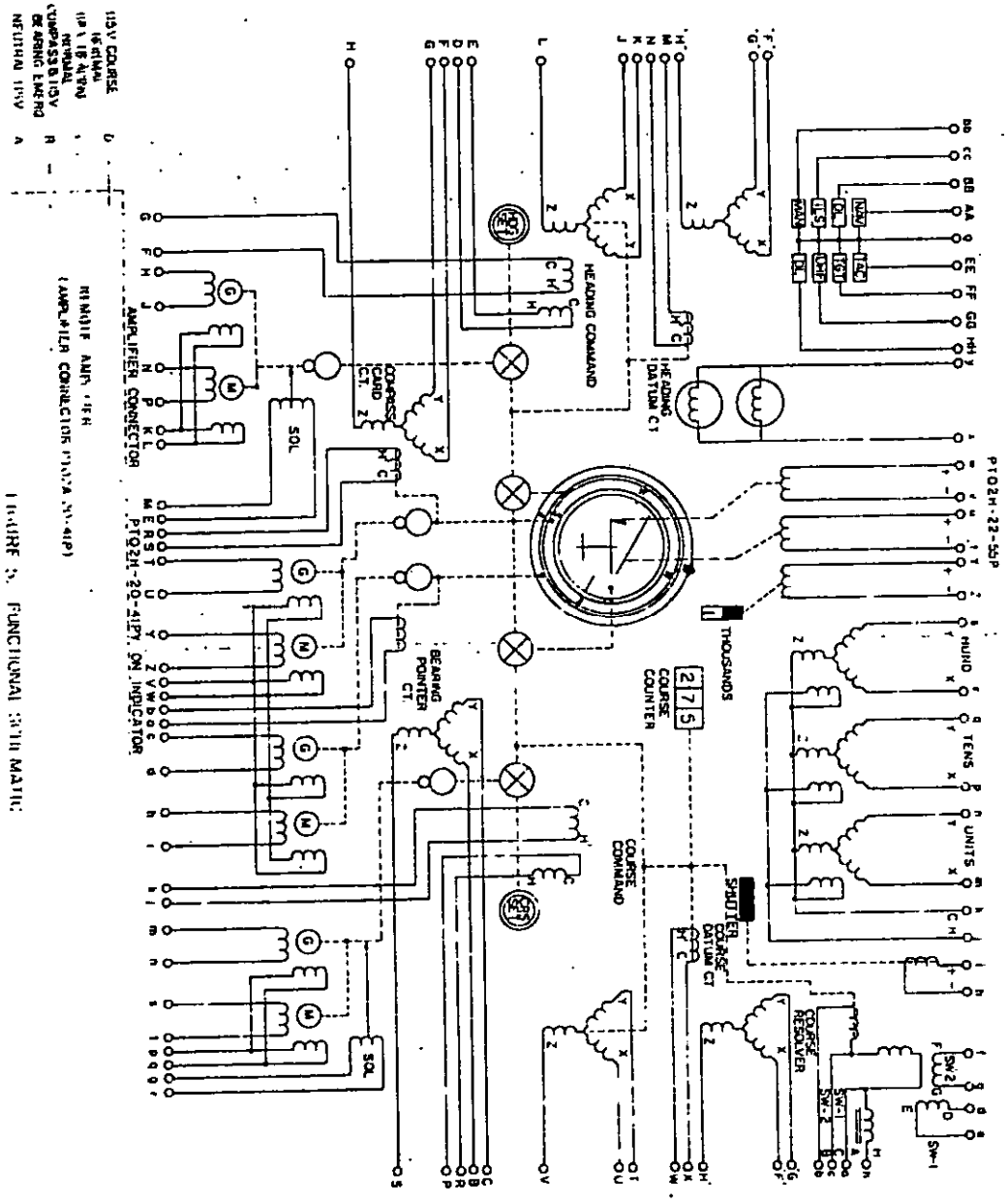


FIGURE 3. FUNCTIONAL SCHEMATIC

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arker shall be between 8 and 10 to 1. The azimuth scale reading ndicated under the heading marker shall be the desired heading. ne heading set knob shall be as shown on figure 2. The maximum acklash between the command heading marker and the azimuth card nall be 1/4

.15.1 HEADING MARKER SERVO SYSTEM. A servo system similar to nat described in 3.14.2 for driving the azimuth ring shall be mployed for position in the heading marker in response to a emote input signal. The servo shall be capable of receiving a hange of command heading at any rate between 0° and 60° per econd without hunting. There shall be no appreciable hunt or scillation in the servo system and it shall position the heading arker around the azimuth ring to within $\pm 1^{\circ}$ of the position of he test synchro. The design of the differential gearing shall esult in operation of the heading marker with respect to the zimuth ring, remote synchro command input and knob command in- ut and knob command input as that achieved on figure 5. The eading marker servo shall conform to the electrical connector nd pin wiring shown on figure 5.

3.15.2 HEADING MARKER SIGNAL INPUT. The heading marker servo ystem shall receive its input signal from a transmitting synchro evice similar to the test transmitter described in 3.11. The eading synchro device shall be a Clifton CDSH-10-AS-4 precision ransolver, or equivalent, which shall operate as a CT and a ransmitter as shown on figure 6. The X, Y, Z loads shall be quivalent to an EP AY500-3 or one exhibiting higher impedance alues.

.15.3 HEADING DATUM CONTROL TRANSFORMER. The heading datum t shall be a Clifton CTC-8-A-4, or equal, and shall operate as hown on figure 5.

.15.4 HEADING COMMAND AND HEADING DATUM CONTROL TRANSFORMER ERO. Both the heading command and heading datum ct's shall be eroed with positive rotation referenced at an index reference f 270°. Both synchros shall be zeroed in the following manner:

.15.4.1 With the standard test transmitter positioned to zero ndication on its dial and locked, and connected to the heading ommand and heading datum ct's as per the X, Y, Z notation indi- ated on figure 5, the standard test transmitter shall be excited ith 26V 400 cycle.

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3.15.4.2 The heading marker shall be positioned to 0° (north) on the azimuth card and the voltage C to H of the heading command ct rotor and heading datum ct rotor as shown on figure 5 shall be null.

3.15.4.3 If the heading marker is rotated cw around the azimuth ring the voltage C to H of the heading command ct rotor and heading datum ct rotor shall be in the opposite phase sense as the voltage C to H of the standard test transmitter from 0 to 180°, null at 180° (south) and shall be in the same phase sense as the voltage C to H of the standard test, transmitter from 180°(south) to 0° (north).

3.15.4.4 If the heading marker is set on 0° (north) on the azimuth dial and if the rotor of the standard test transmitter is rotated to indicate an increasing heading (dial ccw) the voltage C to H of the heading command ct and the heading datum ct shall be null at 0°, in the same phase sense as the excitation voltage C to H from 0° to 180°; null at 180°, and opposite to the phase sense of C to H from 180° to 0°.

3.15.4.5 If the transmitter winding of the heading command ct is excited with 26V ac so that the voltage C to H is in phase with power excitation ground to phase and the heading marker is set on north then the voltage X to Z and Y to Z shall be in phase with C to H and voltage X to Y null. If the heading marker is rotated cw to east the voltage X to Z shall increase before decreasing.

3.15.5 HEADING MARKER SERVO ZERO. The heading command control transformer shall be connected to its servo system. The standard test transmitter shall be connected to the heading command ct as per the X, Y, Z notation on figure 5. The excitation voltage of the test transmitter C to H shall be in phase with the servo power excitation voltage, pin A to pin C on the amplifier connector. The following conditions shall be satisfied.

3.15.5.1 The heading marker shall position to 0° (north) on the azimuth ring when the dial of the standard test transmitter is at 0°.

3.15.5.2 The heading marker shall rotate in an increasing heading indication on the azimuth ring (cw) when the dial of the test transmitter synchro is rotated in an increasing heading indication (dial ccw).

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.15.5.3 When the dial on the standard test transmitter is otated in the increasing heading indication, the voltage C to H f the heading command ct shall be in the same phase sense with he voltage C to H of the standard test transmitter.

.15.5.4 When the transmitter winding of the heading command ct s excited with 26V ac so that the voltage C to H is in phase ith power excitation ground to phase and the heading marker is et on north, the voltage X to Z and Y to Z shall be in phase ith C to H and voltage X to Y null. If the heading marker is otated cw to east, the voltage X to Z shall increase before ecreasing.

.16 COMMAND COURSE. A means of selecting a desired course oth automatically and manually shall be provided. A course arker as shown on figure 2 which can rotate around the inside dge of the azimuth ring shall be driven by a servo system when xcited by a remote device similar to the standard test transitter specified in 3.11 and manually by a knob at the lower ight corner of the instrument. As in the heading marker servo ystem, turning of the course knob shall have no lasting effect n the course arrow when the servo is being driven by a remote evice. The course marker shall rotate in synchronism with the zimuth ring after it is set. The scale reading on the azimuth ing at the tip of the course arrow shall be the selected course r track. The course selection knob shall be connected with the ourse arrow in such a manner that cw rotation of the knob prouces cw rotation of the course arrow, and vice versa. The urning ratio between the knob and the course marker shall be etween 8 and 10 to 1. The course set knob shall be as shown on igure 2. A digital counter shall present the selected course in window on the right side of the indicator as shown on figure 2.

.16.1 COURSE MARKER SERVO SYSTEM. A servo system similar to hat described in 3.14.2 for driving the azimuth ring shall be nployed for positioning the course marker in response to a reote input signal. The servo shall be capable of receiving a hange of command course at any rate between 0 and 60° per secnd without hunting. There shall be no appreciable hunt or scillation in the servo system during steady state and dynamic peration and it shall position the course around the inside dge of the azimuth ring to within ±1° of the position of the est synchro. The design of the differential gearing shall esult in operation of the course marker with respect to the zimuth ring, remote synchro command input and knob command inut as that achieved on figure 5. The course marker servo shall onform to the electrical connector and pin wiring shown on figre 5.

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*3.16.2 COURSE MARKER SIGNAL INPUT. The course marker servo system shall receive its input signal from a transmitting synchro device similar to the test transmitter described in 3.11. The course synchro device shall be a Clifton CDSH-10-AS-4 precision transolver, or equivalent, which shall operate as a CT and a transmitter as shown on figure 6. The X, Y, Z load shall be equivalent to an EP AY500-3 or one exhibiting higher impedance values.

3.16.3 COURSE DATUM CONTROL TRANSFORMER. The course datum ct shall be a Clifton CTC-8-A-4, or equal, and operate as shown on figure 5.

3.16.4 COURSE COMMAND AND COURSE DATUM CONTROL TRANSFORMER ZERO. Both the course command and course datum ct's shall be zeroed as a standard test transmitter at an index reference of 270 Both synchros can be zeroed in the identical manner as the heading command ct and heading datum ct given in 3.15.4 to 3.15.4.4 with course marker substituted for heading marker and all reference made to the course servo system. The course servo system should not be excited with power.

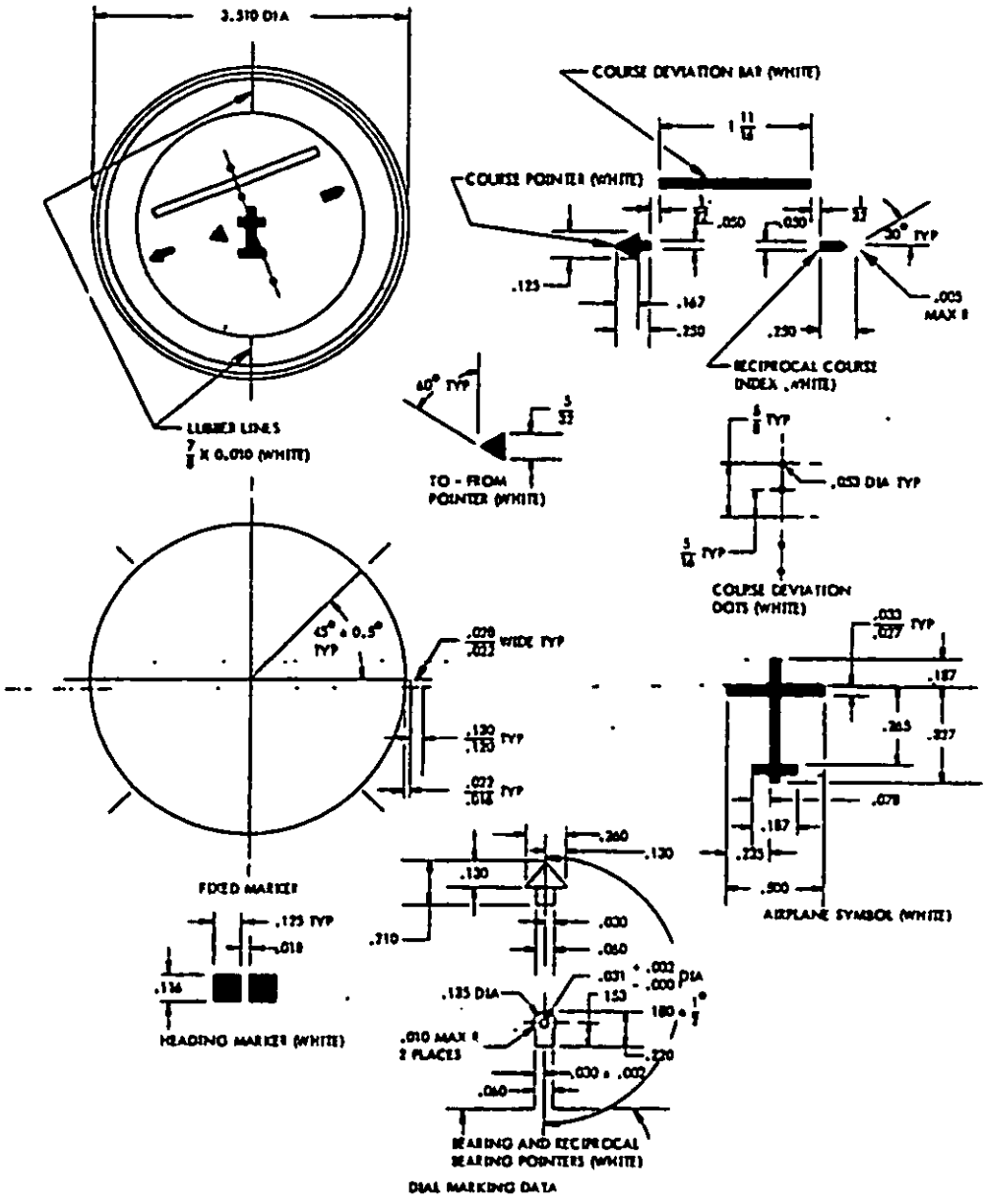
3.16.5 COURSE MARKER SERVO ZERO. The course command control transformer shall be connected to the standard test transmitter as per the X Y Z notation given on figure 5. The excitation voltage of the test transmitter C to H shall be in phase with the servo power excitation voltage, pin A to pin D on the amplifier connector. The following conditions shall be satisfied.

3.16.5.1 The course marker shall position to 0° north on the azimuth ring when the dial of the standard test transmitter is at 0°.

3.16.5.2 The course marker shall rotate in an increasing heading indication on the azimuth ring (cw) when the dial of the test transmitter synchro is rotated in an increasing heading indication (dial ccw).

3.16.5.3 When the dial of the test transmitter synchro is rotated in an increasing heading indication (dial ccw) the voltage C to H of the course command ct shall be in phase with the voltage from C to H of the standard test transmitter.

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LETTERS: 1.155R TO BOTTOM EDGE; .250 HIGH OVERALL; .032-.040 LINE WIDTH.
NUMERALS: 1.217R TO BOTTOM EDGE; .183 HIGH OVERALL; .024-.030 LINE WIDTH.
10° LINES: 1.433R TO INNER END AND EXTENDING TO WITHIN .005 OF O.D. OF PART; .016-.020 LINE WIDTH, LENGTH .125 (REF).
5° LINES: 1.495R TO INNER END AND EXTENDING TO WITHIN .005 OF O.D. OF PART; .008-.010 LINE WIDTH, LENGTH .062 (REF).
NUMERALS AND LETTERS SHALL CONFORM TO MS33559
TOLERANCES: DIMENSIONS IN INCHES, UNLESS OTHERWISE SPECIFIED.
FRACTIONS ± 1/64
DECIMALS ± .005

FIGURE 6. DISPLAY COMPONENT DIMENSIONS

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3.16.5.4 If the transmitter winding of the course command ct is excited with 26V ac so that the voltage C to H is in phase with power excitation ground to phase and the course pointer is set on north, then the voltage X to Z and Y to Z shall be in phase with C to H and voltage X to Y null. If the course pointer is rotated cw to east, the voltage X to Z shall increase before decreasing.

3.16.6 COURSE RESOLVER. A resolver having the electrical characteristics of a Type EP AY221-5-B, or equal, shall be connected to the course gear train to function as shown on figure 5. The resolver shall be electrically connected as shown on figure 5, and zeroed at a scale reference of 0° and an index reference of 300°.

3.16.6.1 Rotor winding number 1 (RW-1) shall be defined as the leading winding that mechanically leads RW-2 by 90° in the direction of higher dial numbers with respect to the azimuth card. At the index reference point (300°) when properly zeroed stator winding 1 (SW-1) and RW-1 shall be directly coupled and SW-2 and RW-2 shall be directly coupled.

3.16.6.2 The command pin of the rotor windings is designated as pin C. The two leads of a stator winding are arbitrarily labeled D and E and this stator winding is called SW-1.

3.16.6.3 The course marker shall be rotated to 300° against the azimuth card. Choose a rotor lead and temporarily label it A. Excite this rotor winding (C being the common ground) with 10V 400 cycle ac power. Adjust resolver rotor shaft (with scale locked at 300°) until voltage D to E of SW-1 is maximum and in phase with voltage A to C, respectively. Lock resolver rotor to scale and rotate scale 90° in a positive direction to 30° against the azimuth card. Remove voltage A to C of RW-1 and apply it to B to C of card. Remove voltage A to C of RW-1 and apply it to B to C of RW-2. Voltage as read across D to E of SW-1 should be maximum and in phase with applied voltage V to C of RW-2. If voltage is out of phase reverse the labeling A and B of the rotor leads and repeat the above zeroing procedure.

3.16.6.4 Set the scale reading to 300° against the azimuth card and excite leads B and C of RW-2. Assign letters F and G to the stator winding SW-2 so that the voltage measured from F to G is in phase with voltage B to C of RW-2.

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16.6.5 Fine zeroing can be accomplished by exciting leads B C of RW-2 and nulling the voltage D to E of SW-1 by rotating solver rotor shaft when scale reading is 300° against the imuth card.

16.6.6 VOR CHARACTERISTICS. The course selector resolver all be equipped with a choke as shown on figure 5 to permit e with VOR systems requiring a 30 cps ORZ (Omni Range Zero). oper ORZ operation at 30 cps shall be obtained when the resolver electrically zeroed with 400 cps excitation at 300°. The ORZ tor leads shall be designated as leads H and C. The contractor all submit the zeroing procedure to the procuring activity for proval prior to fabrication.

16.7 The A to H notation given above shall be connected to nnector PT02H-22-55P, or equal, as specified on figure 5.

NOTE: For 400 cps operation, pins a, b, c, d, e, f, and g are utilized to obtain the sine and cosine relationship. For 30 cps ORZ operation, pins a, c, d, e, f, and g are utilized.

17 COURSE BAR. An indication of the displacement of the aircraft to the right or left of the selected course shall be provided by means of a course bar and dots in accordance with figure

17.1 COURSE BAR MECHANISM. Direct current deflection signals all cause the course bar to move across the face of the instrument. The total movement of the course bar shall be 11/16 ±1/32 ch in both directions measured from the center of the instrument. A movement of 5/8 inch from center shall require 150 ua icroamperes) nominal. A dc polarity of + and - indicated as ns of figure 5 shall cause a deflection to the right when urse arrow is on top. The course bar sensitivity shall be 75 . per dot and range of displacement shall be ±2 dots. The urse bar mechanism shall have a resistance of 1000 ohms ±3 perent and a damping ratio not less than 1.0. The frequency sponse of the course bar mechanism shall be such that amplitude tio of output to input shall not decrease by more than 3 cibels at 0.2 cps. The test limits at standard conditions for e course bar mechanism shall be the following:

±1 dot deflection ±75 ±6 ua
±2 dots deflection ±150 ±12 ua

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3.18 TO-FROM ARROW. A to-from arrow shall be incorporated as shown on figure 2. The to-from arrow shall operate from a meter movement having a resistance of 250 ohms maximum 150 ohms minimum The sensing shall be such that with no signal applied the arrow will not be visible. A signal with polarity as given on figure 5 shall cause the to-from arrow to appear on the same side as the course arrow. A signal with opposite polarity shall cause the to-from arrow to appear on the tail side of the course arrow. A signal not greater than 225 ua shall cause the to-from arrow to reach full displacement.

3.19 The central portion of the display including the course arrow, course bar, the to-from arrow, and course deviation dots shall rotate as a unit turning with the azimuth ring as the airplane's heading changes, in addition to the individual motions listed.

3.20 METER MOVEMENT BEARINGS AND PIVOTS. Meter movements shall turn on hardened ground and highly polished pivot bearings, highly polished first quality sapphire bearings, or other materials that meet the performance requirements of this specification.

3.21 BALANCING. The meter movements shall be balanced by an approved means (threaded nut or splice nuts, or a wire helix moving along a crown arm and counterweight arm) with easily accessible means for rebalancing. The use of solder, shellac, or similar means for holding balance weights will not be acceptable.

*3.22 BEARING MARKER. A bearing marker and reciprocal bearing marker servo system shall be incorporated for positioning the bearing pointer shown on figure 2 from a remote source similar to a standard test transmitter synchro. The servo system shall be similar to the driving azimuth ring. The servo shall be capable of receiving a change in bearing between 0 to 60° per second without hunting. There shall be no appreciable hunt or oscillation in the servo system and it shall position the bearing pointer around the azimuth ring to within ±1° of the position of the test synchro. The servo system shall be connected as specified on figure 5. A single servo loop shall position the bearing and reciprocal bearing pointer.

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3.22.1 BEARING POINTER SIGNAL INPUT. The bearing pointer servo system shall receive its input signal from a transmitting synchro device similar to the test transmitter described in 3.11. The bearing pointer control transformer shall be a Type EP AY500-5, or equal. A size 8 synchro control transformer with the impedance of the stator windings as a Type EP AY-500-5 can be utilized.

*3.22.2 BEARING POINTER CONTROL TRANSFORMER ZERO. The bearing pointer ct shall be zeroes at a scale reference of 180° (under the lubber line at the bottom of the display) and an index reference of 90°. The reciprocal bearing pointer shall be 180° from the bearing marker at all times.

3.22.2.1 With the standard test transmitter connected as per the X, Y, Z notation on figure 5, and excited with voltage from C to H in phase with the servo excitation pins A to B of the amplifier, the transmitter dial shall be positioned to zero.

a. The bearing pointer shall position under the lubber line at the bottom of the display.

b. Rotation of the card on the test transmitting synchro to an increasing heading indication shall cause the bearing pointer to rotate cw.

c. If the rotor of the test transmitter synchro is rotated to an increasing heading indication the voltage C to H of the bearing ct shall be in the same phase sense as the voltage C to H of the standard test transmitter.

3.23 DISTANCE DISPLAY. A digital type distance display shall be provided at the lower left of the instrument as specified on figure 2. The range of the display shall be 000 miles to 1999 miles. The hundreds, tens, and units numerals shall be individually driven by receiver synchros with a 1:1 ratio. The units display shall have 1 mile and 1/2 mile index lines. The numerals shall be centered each 36° of synchro shaft rotation. The receiver synchros shall be Clifton CRC-8-A-1, or equal. The thousand digit shall be driven by a shutter showing black when normally unexcited and 1 when excited with 27V dc.

3.23.1 DISTANCE SYNCHRO ELECTRICAL ZERO. With a standard test transmitter connected as per the X, Y, Z common with C notation and C to H of both the receiver synchros and transmitter synchro excited the three synchros shall be zeroed as follows:

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a. With the test transmitter rotor position set on 0° the 0 shall be in position on the distance dial of each receiver synchro.

b. Rotation of the test transmitter synchro to 36° shall result in 1 being positioned on the distance dial.

c. Increasing numerical dial readings on the receiver synchros shall be indicated for each 36° of the transmitter synchro rotation.

3.23.2 DISTANCE SHUTTER MECHANISM. A black masked shutter shall be provided for obscuring the distance display. The shutter mechanism shall be excited by 27V dc. Voltage excitation shall cause the shutter to disappear from view. The units digit shall be only partially masked by the distance shutter.

*3.24 POWER. The horizontal situation indicator group shall utilize only power which may be supplied by the 115V, 400 cps, single phase ac aircraft electrical system in accordance with MIL-STD-704. Each of the four servo systems shall be excited with 115V, 400 cps on the specific connector and pins specified on figure 5. In addition 5V ac power shall be supplied to the proper connector and pins of figure 5 for operation of the integral lighting and the function words.

*3.24.1 ELECTRICAL POWER VARIATION. The indications shall operate properly and continuously when the ac voltage supplied to the amplifier power connector is in the range between 103 to 124V and at any frequency between 380 and 420 cps or any combination thereof, in addition to transient variation specified in MIL-STD-704.

3.25 CONNECTORS. Two each electrical connectors shall be supplied on the indicator as shown on figure 1. The connectors shall be Scintilla Division, Bendix Aviation Corp., Part No. PT02H-22-55P connector, or equal, and Part No. PT02H-20-41PY, or equal. These connectors shall mate with Part No. PT06A-22-558 and PT06A-20-41SY, respectively.

3.25.1 The amplifier connector shall be Scintilla Part No. PT02A-20-41P, or equivalent. The connections between indicator and amplifier shall be as shown on figures 4 and 5.

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*3.26 WEIGHT. The total weight of the indicator and amplifier shall not exceed 12.2 pounds. The weight of the amplifier shall not exceed 4.2 pounds.

3.27 SCREW THREADS. Unless otherwise specified, the threads of all machine screws 0.060, or larger, in diameter shall conform to MIL-S-7742.

*3.28 LUBRICATION. The horizontal situation indicator group shall be lubricated in accordance with MIL-STD-838.

3.29 FINISHES AND PROTECTIVE COATINGS

3.29.1 ALUMINUM ALLOY PARTS. Aluminum parts shall be covered with an anodic film conforming to MIL-A-8626, except as follows:

3.29.1.1 Dials, small holes, and case inserts need not be anodized.

3.29.1.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.29.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.29.1.4 Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.29.1.5 When abrasion resistance is a factor, chemical film in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.29.2 CADMIUM PLATING. Cadmium plating shall be in accordance with QQ-P-416, type I, class C.

3.30 IDENTIFICATION OF PRODUCT. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130 and as follows:

Indicator or Amplifier (as applicable)
Component of
Horizontal Situation Indicator Group AF/A24J-1

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3.31 INSTALLATION

3.31.1 MOUNTING SCREWS. The contractor shall furnish sufficient mounting screws for installing the horizontal situation indicator group. They shall be round head brass machine screws and shall have a durable dull black finish. Length shall be sufficient to permit mounting on a panel 3/16 inch thick. Screw threads shall be size 10-32NF-2.

3.31.2 ENVELOPE. An envelope, containing the mounting screws, shall be furnished with each indicator. The following information shall be printed on the face of the envelope:

IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS

3.32 WORKMANSHIP

3.32.1 GENERAL. The horizontal situation indicator group, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws, and bolts.

3.32.2 RIVETING. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.32.3 CLEANING. The horizontal situation indicator group shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips, and other foreign materials removed during and after final assembly.

3.32.4 DIMENSIONS. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the horizontal situation indicator group they shall be held or limited accordingly.

3.32.5 SCREW ASSEMBLIES. Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt, or threads.

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4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFICATION OF TESTS. The inspection and testing of the horizontal situation indicator group shall be classified as follows:

a. Acceptance tests. See 4.2.

b. Qualification tests. See 4.5.

4.2 ACCEPTANCE TESTS. Acceptance tests shall consist of:

a. Individual tests

b. Sampling plans and tests

*4.2.1 INDIVIDUAL TESTS. Each horizontal situation indicator group shall be subjected to the following tests in the specified sequence and as described under 4.4 TEST METHODS:

a. Examination of product

b. Synchro and resolver zero

c. Servo operation

d. Servo sensitivity

e. Scale error

f. Servo follow-up rate and damping

g. Cause bar mechanism (accuracy)

h. Balance

i. Accuracy

j. Stop

k. To-From arrow

l. Distance display

m. Shutter

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n. Sealing

o. Fogging

p. Control knobs.

4.2.2 SAMPLING PLANS AND TESTS

4.2.2.1 SAMPLING PLAN A. Horizontal situation indicator group selected in accordance with the following schedule shall be subjected to the following tests in the specified sequence and as described under 4.4 TEST METHODS:

Number to be Procured	Subjected to Sampling Plan A Tests
First 10	1
Next 90	2
Each additional 100	1

a. Individual tests

b. Power consumption

c. Magnetic property

d. Acceleration

e. Dielectric

f. Vibration error

g. Servo null position shift

h. Low temperature operation

i. High temperature operation

j. Course bar mechanism (dynamic)

k. Coatings.

*4.2.2.2 SAMPLING PLAN B. Unless otherwise specified by the procuring activity, 3 horizontal situation indicator groups selected at random from the first 15 horizontal situation indicator groups of the contract or order shall be subjected to the following tests in the specified sequence and as described under 4.4 TEST METHODS:

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a. Sampling plan A tests

b. Electromagnetic interference

c. High temperature exposure

d. High altitude, low temperature

e. Temperature shock

f. Humidity

g. Fungus

h. Salt spray

i. Sand and dust

j. Vibration failure

k. Acceleration

l. Life

m. Sealing thermal shock

n. Cover glass abrasion.

*4.2.2.2.1 At the option of the procuring activity the tests may be divided into three groups as follows, to be conducted on three sets of equipment, if a need exists to expedite tests. Each indicator and amplifier shall meet the requirements of all the individual tests after completing the group of specific tests to which submitted. At the option of the procuring activity, group II may be conducted on empty cases which have all external parts installed or attached as on a completed indicator and amplifier assembly case.

Group I	Group II	Group III
Radio noise interference	Humidity	Vibration failure
High temperature exposure	Fungus	Acceleration
Low temperature	Salt spray	Life
Temperature shock	Sand and dust	
	Sealing thermal shock	
	Cover glass abrasion	

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4.2.2.3 REJECTION AND RETEST. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.2.2.3.1 INDIVIDUAL TESTS MAY CONTINUE. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.2.3 DEFECTS IN ITEMS ALREADY ACCEPTED. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.3 TEST CONDITIONS

4.3.1 STANDARD ATMOSPHERIC CONDITIONS. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at room pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C).

4.3.2 VOLTAGE-INPUT. Unless otherwise specified the display shall be tested with 115V 400 cps (nominal) single phase power connected to the specific pins of the amplifier connector.

4.3.3 PROPERLY CONNECTED. When the display is referred to as being properly connected, it shall be understood that it is connected to a test fixture that is providing a lighting circuit voltage variable from 5V to approximately 1V, synchro signals for remote inputs to the servo systems, variable current dc signals with reversible polarity to the meter movements, 27V dc for shutter excitation, and ac power as required for operation of the servo systems.

4.3.4 VISUAL OBSERVATION. Unless otherwise specified, all visual observations of indicator performance shall be made at a distance of 2 feet.

4.4 TEST METHODS

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4.4.1 EXAMINATION OF PRODUCT. Each horizontal situation indicator group shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.4.2 SYNCHRO AND RESOLVER ZERO. Each transolver ct receiver synchro and resolver shall be checked for zero by the method specified in section 3.

4.4.3 SERVO OPERATION. The dynamic accuracy of the four servos shall be determined in the following manner: The rotor of an identical ct, with a locked shaft not connected mechanically to the servo gear train shall be wired directly into the servo amplifier. The stator leads of this external synchro shall then be connected to the test synchro transmitter, forming an "open loop" servo system. The test transmitter (or external synchro) shall be rotated until a null signal is obtained on the rotor of the external synchro, stopping rotation of the servoed dial or pointer. Error signals corresponding to various dynamic errors shall then be introduced into this open loop servo by turning the test transmitter to positions differing from the null position by 1°, 2°, 3°, and 4°. At each setting, the resulting speed of dial or pointer rotation shall be measured.

*4.4.3.1 A decrease of 15 percent in minimum servo speed is acceptable during the high and low temperature test of 4.4.17, 4.4.18, and 4.4.22.1. Each of the servos shall meet the following requirements:

Synchro Error Signal Open Loop Input	Minimum Indicator Speed (Output)
1	20°/sec
2	30°/sec
3	45°/sec
4	60°/sec

4.4.3.2 SERVO SENSITIVITY. The indicator shall be properly connected. The rotor of the calibrated synchro shall be rotated 1° and there shall be a perceptible movement in each of the pointers and dial as their respective ct's are excited.

*4.4.3.3 SCALE ERROR. The indicator shall be properly connected. The standard calibrated synchro shall be set on zero and the heading indication read and recorded. The calibration synchro connected to the azimuth card shall be turned successively to headings of 30°, 60°, 90°, 120°, 150°, 180°, 210°, 240°,

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270°, 300°, 330°, and 360° and the readings under the lubber line recorded. The azimuth card readings shall not differ from the corresponding synchro heading by more than ±0.5° in at least 9 headings and ±0.8° in the remaining 3 tested. At end of paragraph, add: "In any sampling plan A or B test which requires repeating the scale error test, the azimuth card tolerance shall be ±1.0 at each heading." The identical procedure will be followed for the heading and course marker except that the reading shall be with respect to the azimuth card, and any one reading shall not differ from the transmitter indication by more than ±1°. The identical proceudre will be followed for the bearing pointer except that the test transmitter shall be set at 180° (bearing pointer will position at the top lubber line) and readings taken relative to this zero. The bearing pointer error at any one heading shall not exceed ±1°

4.4.4 SERVO FOLLOW-UP RATE AND DAMPING. The indicator shall be properly connected. The calibrated synchro connected to the azimuth card shall be de-energized, its rotor rotated 90° and re-energized. The time required for the azimuth ring to rotate between 5°and 85° of its original displacement from the calibrated synchro shall not exceed 1.5 seconds, and the total time before the azimuth ring comes to rest shall not exceed 2 seconds The identical test shall be performed on each of the other servo systems.

4.4.5 COURSE BAR MECHANISM (ACCURACY). The course bar mechanis shall operate with the polarity and sensitivity specified in 3.17.1.

4.4.5.1 BALANCE TEST. A static balance test with the normal 1g force of gravity acting successively along each of the mutually perpendicular axis of the case shall not cause the bar to deviat from its center position more than 1/2 bar width.

*4.4.5.2 ACCURACY TEST. An accuracy test with plus or minus direct current applied to the proper terminals shall result in the following:

±1 dot deflection	±75 ua ±6 ua
±2 dots deflection	±150 ua ±12 ua

The tolerances for the high and low temperature tests specified in 4.4.17, 4.4.18, and 4.4.22.1 shall be as follows:

±1 dot deflection	±75 ±10 ua
±2 dots deflection	±150 ±17 ua

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4.5.3 STOP TEST. Twice normal current shall be applied in ch direction and slowly reduced. The pointer shall not stick.

4.6 TO-FROM ARROW. The sensitivity of the To-From arrow .all be checked by applying specified current in each direction specified in 3.18. Double voltage shall be applied in both .rections and slowly reduced. There shall be no sticking.

..4.7 DISTANCE DISPLAY. Standard test transmitter synchros ;all be applied to each receiver synchro. A scale error test ;all be conducted for each numeral of the units, tens, and ;ndreds counters. Each numeral shall center for each 36° $+4^{\circ}$;crement of the test transmitter as specified in 3.23.1. A ·ltage of 27V dc shall be applied to the thousands shutter and ;e thousands digit shall appear. Counter motion shall be smooth .th no sticking.

4.8 SHUTTER. Proper polarity of 28V dc shall be applied to ;e shutter mechanism. The shutter shall disappear from view.

4.9 SEALING. The indicator shall be tested for leaks by means a mass spectrometer type of helium leak detector. The initial ;ximum detected leak rate at a pressure differential of 1 atmos-;ere shall not permit more than 10 percent loss of the total .lling medium after 1000 hours.

..4.9.1 FOGGING. The indicator shall be energized and placed ; a $+71^{\circ}$ $+2^{\circ}$ C controlled ambient for a minimum period of 1 ·ur. At the end of this period and while still at this tempera-;re, an ice cube shall be rubbed on the indicator glass face ·r a period of from 1 to 2 minutes. The glass shall be wiped ·y (do not use compressed air) and the indicator inspected for ·idence of water or oil fog. Evidence of fogging shall be ;use for rejection.

;.4.10 POWER CONSUMPTION. The power consumption of the indica-·r and amplifiers shall be measured with suitable meters. The : consumption excluding mode and integral lighting shall not ;ceed 45 va with all servos operating simultaneously.

4.11 The indicator and amplifier shall be properly connected. ;e ac voltage shall be reduced to 103V and 380 cps. The system ;all meet the requirements specified in the individual tests.

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*4.4.12 MAGNETIC PROPERTY. This test shall be made first with no power applied to the indicator and then shall be repeated with the indicator operating on rated power. The indicator shall be held in various positions magnetically east or west of and 12 inches from the center of the free magnet approximately 1-1/2 inches long in a magnetic field with a horizontal intensity of 0.18 ±0.01 oersted, and the maximum deflection of the magnet shall not exceed 5°. An aircraft compass with the compensating magnets removed may be used as the free magnet for this test.

4.4.13 ACCELERATION TESTS. This indicator shall be properly connected and power applied to the servos and meter movements. The indicator shall be accelerated along each axis with a force of 5g. The servoed pointers and azimuth card shall not vary by more than 1° from the initial indication and the deviation bar shall not vary by more than 1 bar width from its initial position. The To-From arrow with no power applied shall not appear into view.

4.4.14 DIELECTRIC. A potential of 500V dc shall be applied between isolated pins and between pins and the case for a period of 10 seconds. There shall be no breakdown of insulation.

4.4.15 VIBRATION ERROR. The indicator shall be mounted as specified in procedure IV of MIL-E-5272, properly connected and shall be subjected to vibration with a total amplitude between 0.003 and 0.005 inch and shall meet the following requirements.

4.4.15.1 The maximum total amplitude of pointer oscillation, azimuth card oscillation, and deviation bar oscillation shall not exceed 1/2° in pointer and azimuth or 1/2 of the deviation bar width.

4.4.15.2 The maximum variation of the pointers azimuth card, and deviation bar during vibration from the original indication before vibration throughout the frequency range, shall be noted and shall not exceed 1/2° in pointer and azimuth card or 1/2 of the deviation bar width.

4.4.15.3 Oscillation and variation tests may be conducted at any indication at the discretion of the inspector.

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4.15.4 The indicator shall be properly connected to the ampli- er with power applied. The amplifier shall be subjected to a bration error test consisting of a frequency survey with vibra- on applied to longitudinal, lateral, and vertical axis of the plifier. The amplifier shall be subjected to vibration with a nstant applied double amplitude of 0.060 inch through the fre- ency range of 10 to 55 cps and 10g input from 55 to 500 cps in ch of the 3 mutually perpendicular axis. This test may be con- cted with vibration applied in a circular motion in a plane lted 45° to the horizontal plane of the amplifier. The dura- on of the vibration shall be a minimum of 2 minutes in each is for 1 complete cycle of frequency range or a minimum of 2 nutes with the vibration applied in a circular motion for 1 mplete frequency range. There shall be no transients or dis- ntinuites in the outputs during the vibration. All servos shall erate normally throughout the test.

.4.16 SERVO NULL POSITION SHIFT. The four servos shall be operly connected to standard test transmitters set at the zero dication. The input voltage and frequency shall be varied thin the following limits and combinations:

Frequency	380 to 420 cps
AC voltage	103 to 127V

e zero indication of the servoed pointers and azimuth card all not shift more than 1° from the normal null position.

4.17 LOW TEMPERATURE OPERATION. The horizontal situation dicator group shall be properly connected, except that no wer shall be applied, and shall be subjected to a temperature -54° ±2°C for a period of 4 hours. At the end of this period d while still at the low temperature, power shall be applied. ere shall be no sticking of movable component parts, and inters shall move throughout their range of travel without icking. There shall be no failure as a result of this test. ere shall be a 5-minute warmup period after the application of wer before data is taken. The indicator shall meet the require- nts of 4.4.3, 4.4.5, 4.4.6, 4.4.7, and 4.4.8.

4.18 HIGH TEMPERATURE OPERATION. The horizontal situation dicator group shall be properly connected with power applied, d shall be placed in a chamber and maintained at a temperature +71° ±2°C for a period of 4 hours. There shall be no sticking

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of movable component parts, and pointers shall move throughout their range of travel without sticking. There shall be no failure as a result of this test. The indicator shall meet the requirements of 4.4.3, 4.4.5, 4.4.6, 4.4.7 and 4.4.8.

4.4.19 COURSE BAR MECHANISM (DYNAMIC). The course bar mechanism shall meet the dynamic response characteristics specified in 3.17.1.

4.4.20 COATINGS. The cover glass and wedge shall by evaluated to determine transmittance and reflectance of light emitted from a 2854°K source. These values shall not exceed those specified in section 3.

*4.4.21 ELECTROMAGNETIC INTERFERENCE. The electromagnetic interference test shall be conducted in accordance with MIL-STD-462 for class ID equipment.

4.4.22 ENVIRONMENTAL. The horizontal situation indicator group shall be subjected to the following tests conducted in accordance with the specified procedures of MIL-E-5272 and as specified herein.

4.4.22.1 HIGH TEMPERATURE EXPOSURE. Procedure I for a period of 24 hours and properly connected, except that no power shall be applied. At the end of this period and while still at the high temperature, the indicator and amplifier shall meet the requirements specified in 4.4.4 through 4.4.8. The horizontal situation indicator group shall then undergo a temperature altitude combination test in accordance with MIL-E-5400 for class I equipment. Intermittent operation shall consist of a 30-minute test. The indicator shall operate without sticking or erratic performance during the test.

4.4.22.2 HIGH ALTITUDE, LOW TEMPERATURE. Low temperature tests procedure II, except that examination shall be performed only at the end of the 72-hour period. The horizontal situation indicator group shall be properly connected, except that no power shall be applied. At the end of this period, the pressure in the chamber shall be reduced to the equivalent of 80,000 ±500 feet and power applied. There shall be no binding or sticking at any possible indication. After the horizontal situation indicator group has returned to room temperature, it shall meet all individual tests. There shall be a 5-minute warmup period after the application of power before data is taken.

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.22.3 TEMPERATURE SHOCK. Procedure I. After the test the izontal situation indicator group shall meet the requirements all individual tests.

.22.4 HUMIDITY. Procedure I. The horizontal situation indi-or group shall then meet the requirements specified in 4.4.4 ough 4.4.8. There shall be no evidence of corrosion or ting of the internal or external parts of the horizontal situ-on indicator group which would adversely affect subsequent ration.

.22.5 FUNGUS. Procedure I. After this test the horizontal uation indicator group shall meet the requirements specified 4.4.4 through 4.4.8. There shall be no deterioration which ld adversely affect subsequent operation and fungus growth ll not be supported.

.22.6 SALT SPRAY. Procedure I for a period of 50 hours. At end of the 50-hour period, the horizontal situation indicator up shall meet the requirements specified in 4.4.4 through .8. No damage which would adversely affect subsequent opera-n shall result from this test, and the tolerances specified er the individual tests shall apply.

.22.7 SAND AND DUST. Procedure I. At the end of this test, horizontal situation indicator group shall be removed and owed to cool at room temperature and shall then meet the uirements specified in 4.4.4 through 4.4.8.

.22.8 VIBRATION FAILURE. The indicator shall be subjected to ibration test in accordance with procedure V. The amplifier ll be subjected to procedure I. Following the vibration test, horizontal situation indicator group shall meet the require-ts specified in 4.4.4 through 4.4.8. No looseness in the hanism nor damage to any part of the horizontal situation icator group shall result from the vibration.

.23 ACCELERATION. The horizontal situation indicator group operating shall be mounted in its normal operating position accelerated first along its vertical axis and then along h of 2 axes that are perpendicular to the vertical axis and each other. The indicator shall be subjected to an accelera-n of 20g in each of the above axes for a period of 1 minute. the end of this acceleration the indicator shall meet the uirements for the individual tests. No damage to the horizon-situation indicator group shall result from this test.

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4.4.24 LIFE. The horizontal situation indicator group shall be properly connected, and power supplied to each of the 4 servo systems and the lighting circuit. The test shall last for 1000 hours. The 4 servos and 2 distance counters shall be operated at 20° per second for approximately 720° and then alternately in the opposite direction. The course bar and to-from arrow shall be driven through their range 5 times per minute. The distance shutter shall be operated for 30 minutes for every hour period. The heading and course knobs shall be rotated for 20 revolutions in each direction every 30 minutes with the remote transmitting synchro source unexcited. There shall be no failure during the test. After the test, the indicator shall meet the requirements specified for individual tests.

4.4.24.1 Failures that occur shall be evaluated to determine whether they are purely random or whether they are systematic due to a design or component weakness. Suitable corrective action must then be determined and action taken accordingly. The procuring activity shall be notified of all failures occuring during this test and results of the investigation.

4.4.25 SEALING THERMAL SHOCK. The indicator shall be immersed alternately in tap water maintained at 85°C +5°C and 5°C +4°C for a total of 8 cycles. The length of time for each bath immersion shall be 30 minutes; not more than 5 seconds shall elapse between bath immersions. No damage to the hermetic seal shall result at the completion of this test. Following the immersions, the indicator shall be tested for leaks by means of a mass spectrometer type of helium leak detector in accordance with MIL-D-4815. The initial maximum detected leak rate at a pressure differential of 1 atmosphere shall not permit more than 10 percent loss of the total filling medium after 1000 hours.

*4.4.26 COVER GLASS ABRASION. An abrasive pad shall be pressed firmly against and pulled across the cover glass in two different directions in the form of a cross. The glass shall be scratched once in each direction. Pressure shall be applied gradually within the case until a 20 psi differential exists. This pressure shall be maintained for 30 minutes. There shall be no damage as a result of this test. The abrasive pad shall be 600-grit sandpaper, 3/4 inch wide by 3 inches long mounted on a wooden base.

*4.4.27 CONTROL KNOBS. The torque required to rotate the course and heading set knobs shall be measured throughout one revolution of the course arrow and one revolution of the heading marker in both clockwise and counterclockwise directions. The torque required shall be within 2 to 10 ounce-inches.

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5 QUALIFICATION TESTING

5.1 QUALIFICATION TEST SAMPLES. The qualification test mples shall consist of 3 horizontal situation indicator groups. mples shall be identified with the manufacturer's own part nber and any additional information required by the letter of :horization.

5.2 QUALIFICATION REQUIRED. Prior to actual procurement, the >duct which this specification covers shall pass the qualifica->n tests specified herein. If the product is later modified in y way, the modified form shall be subjected to and shall pass e same qualification tests.

5.3 QUALIFICATION TESTS. The qualification tests shall con-st of all the tests specified herein.

.6 PACKAGING INSPECTION

.6.1 QUALITY CONFORMANCE. The inspection of the preservation-:kaging and interior package marking shall be in accordance th group A and B quality conformance inspection requirements, :tion 4 of MIL-P-116. The sampling and inspection of the :king and marking for shipment and storage shall be in accord-:e with the quality assurance provisions of applicable con-iner specification and the marking requirements of MIL-STD-129.

.6.2 FIRST ARTICLE. When the unit container is capable of rving as the shipping container First Article Inspection, and ugh handling tests as outlined in Section 4 of MIL-P-116, all be accomplished followed by a functional test of the unit insure freedom from operational malfunction.

PACKAGING

.1 PRESERVATION-PACKAGING. Preservation-packaging shall be vel A, B, or C as specified by procuring activity.

.1.1 LEVEL A.

.1.1.1 CLEANING. Unit shall be cleaned in accordance with ocess C-1 of MIL-P-116.

.1.1.2 DRYING. Immediately after cleaning, the unit shall be ied following any one or combination of the drying procedures MIL-P-116. The drying procedures employed shall not be jurious to the unit.

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*5.1.1.3 PRESERVATION APPLICATION. Not required.

*5.1.1.4 UNIT PACKAGING. Unless otherwise specified by the procuring activity, each unit shall be packaged in quantity unit packs of one each in accordance with Method IA8 of MIL-P-116. Unit container to be PPP-B-636 weather-resistant fiberboard. Apply sufficient cushioning material between primary and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the unit, when completed packs are subjected to rough handling drop tests of MIL-P-116. Insofar as practical, exterior containers shall be of uniform shape, minimum tare and cube consistant with protection required.

*5.1.2 LEVEL B. Each unit shall be packaged as indicated for Level A except that a specification PPP-B-636 class domestic shall be used as the unit container.

*5.1.3 LEVEL C. Units shall be cleaned, dried, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

*5.2 PACKING. Packing shall be level A, B, or C as specified (see 6.2).

*5.2.1 LEVEL A. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636 weather-resistant. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636. For the Army, specification PPP-B-601 plywood box, shall be used for the shipping container.

*5.2.2 LEVEL B. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636, class domestic. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636. For the Army, the shipping containers shall conform to specification PPP-B-636 class weather resistant.

*5.2.3 LEVEL C. Packaged units shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to applicable carrier rules and regulations.

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5.3 MARKING OF SHIPMENTS. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

Indicator or Amplifier (as applicable) Component of Horizontal Situation Indicator Group AF/A24J-1

5.3.1 REINSPECTION DATE. The reinspection date markings shall be in accordance with the instructions of the procuring activity.

6. NOTES

6.1 USE. The AF/A24J-1 horizontal situation indicator group is intended for use in high performance aircraft as a component of flight director system. The display will be controlled by reference and remote command sources providing transmitter synchro output signals and dc meter movement signals.

6.2 ORDERING DATA. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Level of packaging and packing desired

c. Reinspection date of horizontal situation indicator group's in storage

d. Whether sampling plan B tests are to be conducted.

6.3 QUALIFICATION. With respect to products requiring qualification, awards will be made only for such products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is HQ AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016, and information pertaining to qualification of products may be obtained from that activity.

6.4 DEFINITIONS

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6.4.1 INDEX REFERENCE POSITION. The index reference position of a synchro is that position specified in terms of dial reading where the voltage XY is zero and the stator voltage XZ and YZ is 180° out of phase with the rotor voltage HC. When the synchro is in this position, the voltage between H and X will be the sum of the rotor and stator voltages. The index reference position of the standard test transmitter is 0°; of the distance counter receiver synchros it is 0°; of the control transformers in course and heading it is 270°; and the bearing pointer control transformer is 90°.

6.4.2 HERMETIC SEAL. A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic materials or both. A hermetic seal is not intended to include seals accomplished by gaskets.

*6.5 CHANGES FROM PREVIOUS ISSUE. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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